

01-11B LUBRICATION [WL-C, WE-C]**LUBRICATION SYSTEM OUTLINE**

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LUBRICATION SYSTEM

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LUBRICATION SYSTEM OUTLINE [WL-C, WE-C]

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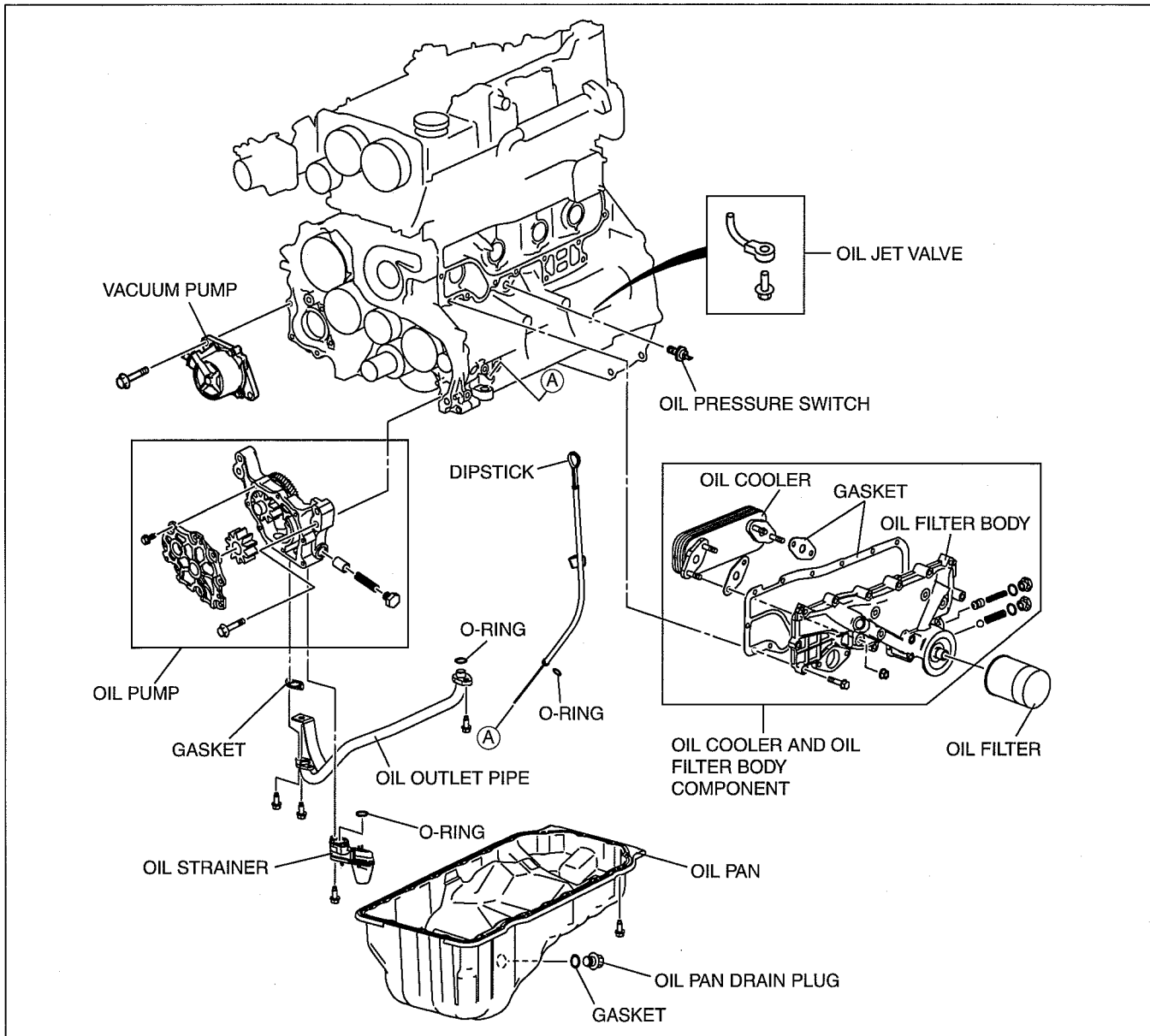
Features

Improved lubricity	• Gear type oil pump adopted • Oil jet valves adopted • Water-cooled type oil cooler adopted
Reduced weight	• Plastic oil strainer adopted
Reduced engine noise	• Oil pan baffle plate adopted

LUBRICATION [WL-C, WE-C]

LUBRICATION SYSTEM STRUCTURAL VIEW [WL-C, WE-C]

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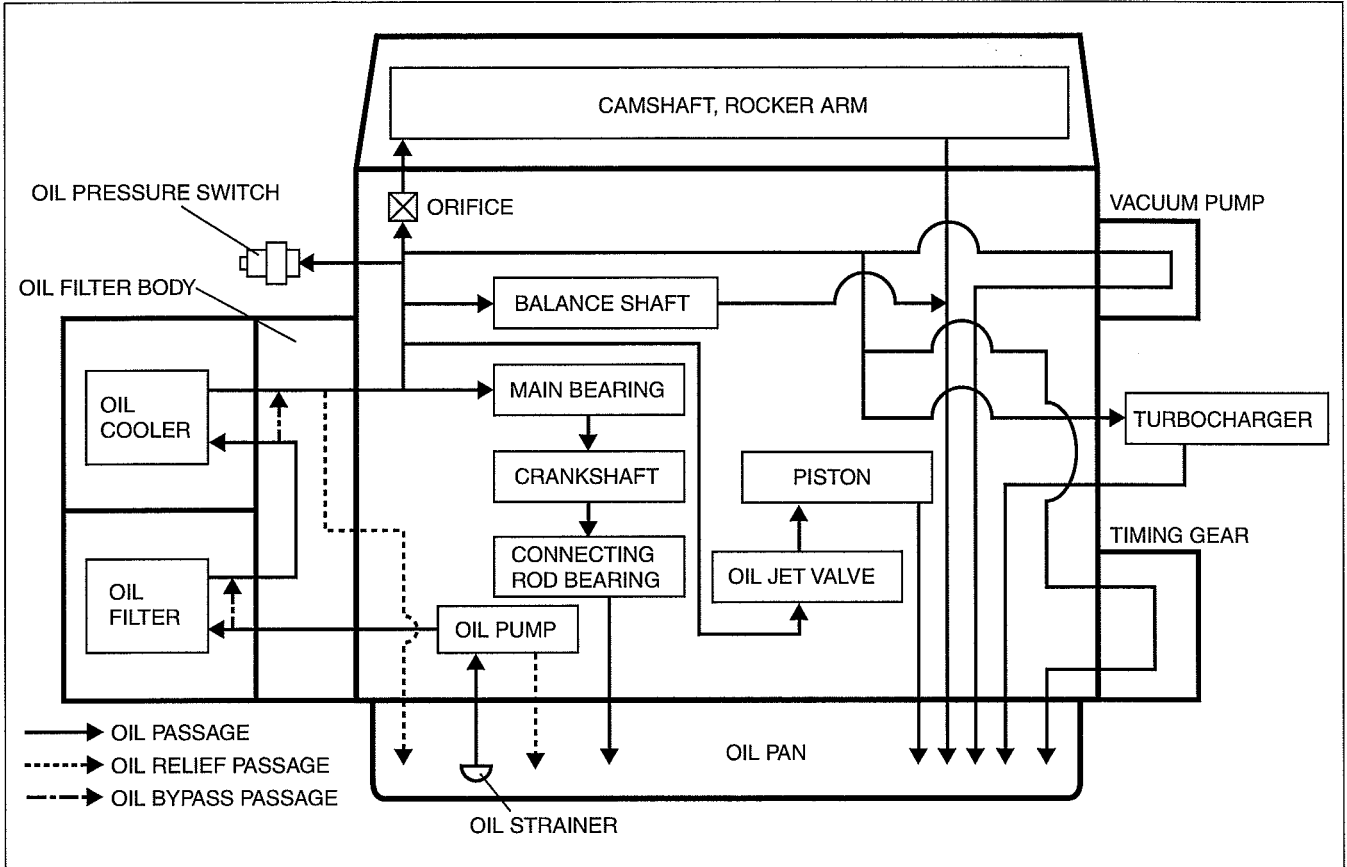


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LUBRICATION [WL-C, WE-C]

LUBRICATION SYSTEM FLOW DIAGRAM [WL-C, WE-C]

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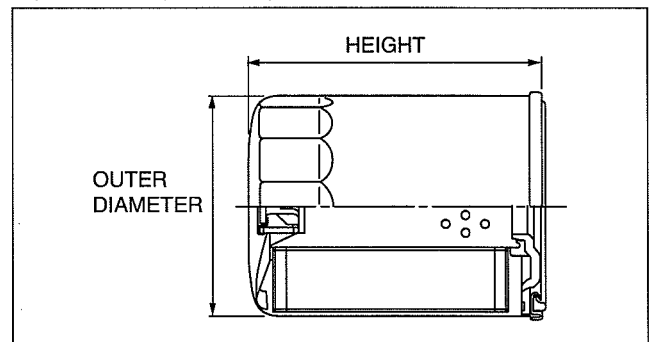


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OIL FILTER CONSTRUCTION [WL-C, WE-C]

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- A full-flow paper element type oil filter has been adopted.
- The oil filter is installed on the oil filter body. (Left surface (vehicle left) of the cylinder block)



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Oil filter specification

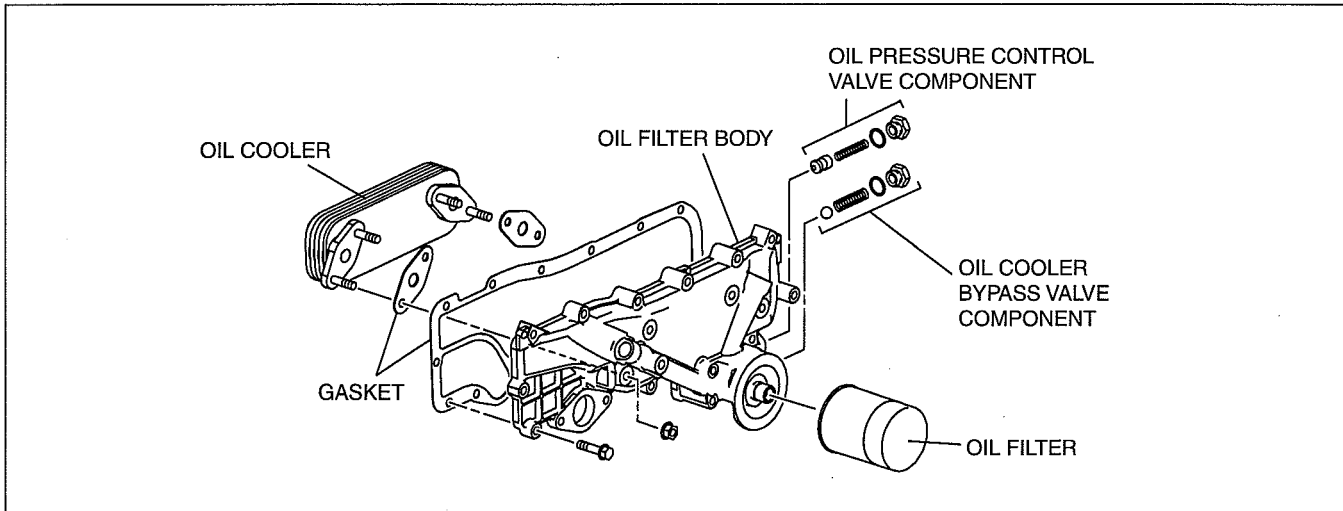
Item	Specification (mm {in})
Outer diameter	73.7 {2.90}
Height	100 {3.94}

LUBRICATION [WL-C, WE-C]

OIL COOLER, OIL FILTER BODY CONSTRUCTION [WL-C, WE-C]

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- A water-cooled, 5-layered type oil cooler has been adopted to reduce engine oil degradation.
- The oil cooler, oil cooler bypass valve, and oil pressure control valve are built into the oil filter body.



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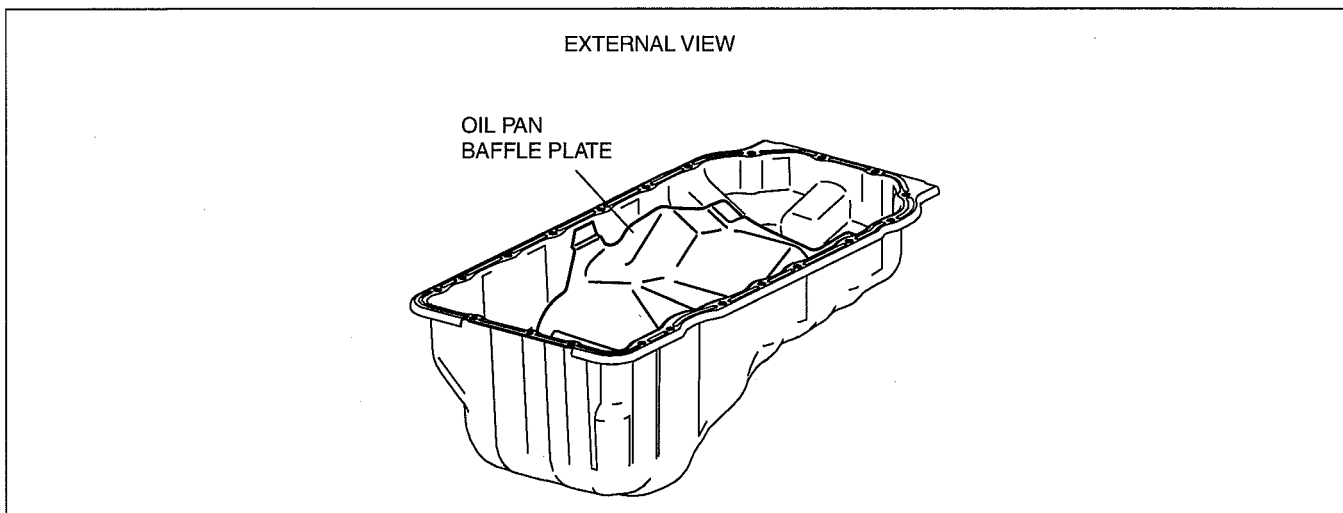
Oil filter body specification

Item	Specification (kPa {kgf/cm ² , psi}) [oil temperature: 120 °C {248 °F}]
Oil cooler bypass valve opening pressure	164—200
Oil pressure control valve opening pressure	400—480

OIL PAN CONSTRUCTION [WL-C, WE-C]

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- A steel oil pan has been adopted.
- A silicon sealant with excellent sealing qualities has been adopted. Also, sealing slots have been adopted on the oil pan attachment side to improve sealing performance.
- An oil baffle plate has been installed inside the oil pan to bifurcate the oil pan bottom. Due to this, the vibrations at the oil surface and in the oil pan have been reduced for improved engine noise suppression.



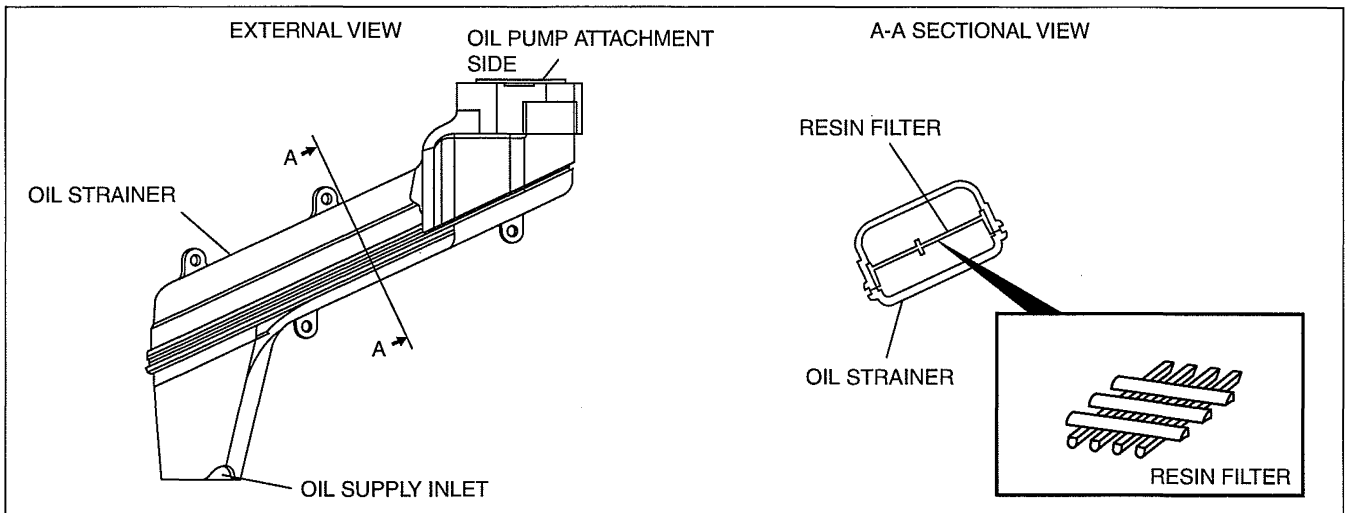
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LUBRICATION [WL-C, WE-C]

OIL STRAINER CONSTRUCTION [WL-C, WE-C]

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- A plastic oil strainer with a resin filter in the middle of the strainer has been adopted for weight reduction.

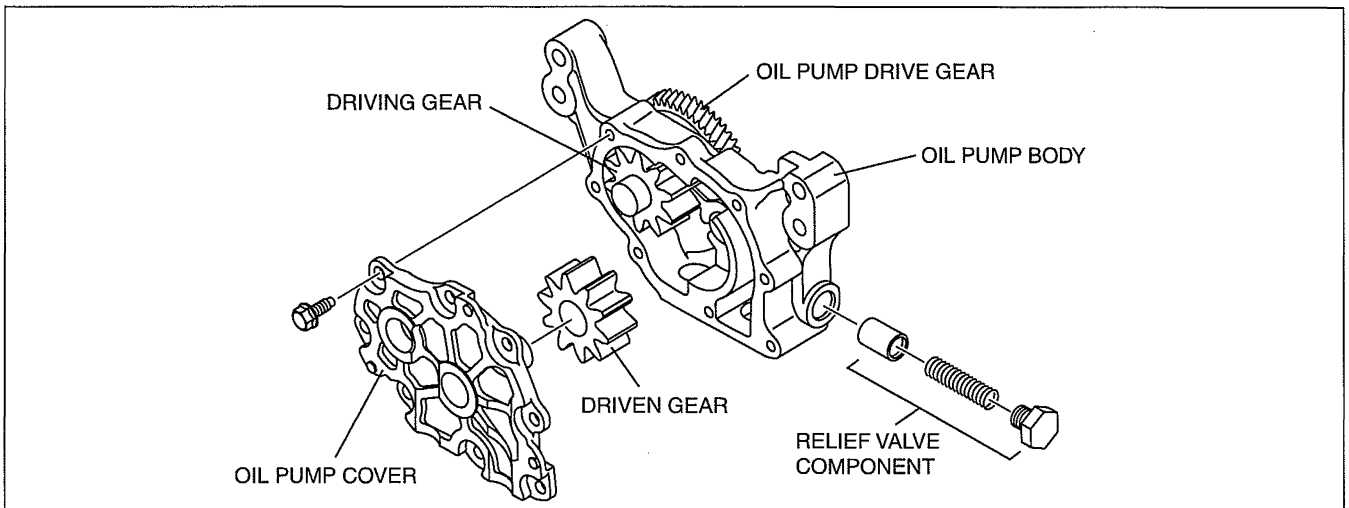


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OIL PUMP CONSTRUCTION [WL-C, WE-C]

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- A gear type oil pump with superior heat resistance and noise reduction has been adopted, ensuring stable discharge performance.
- The oil pump consists of the oil pump body, oil pump drive gear, driving gear, driven gear, relief valve, and oil pump cover.
- The oil pump is driven by the crankshaft through the timing gear and oil pump drive gear.
- The relief valve is built into the oil pump body.



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Oil pump specification

Item	Engine speed [rpm]	Specification (kPa {kgf/cm ² , psi})
Oil discharge pressure (reference value) [after warm up]	At idling	100—330 {1.02—3.36, 14.5—47.7}
	2,500	410—570 {4.19—5.81, 59.6—82.6}
Relief valve opening pressure (reference value)		580—700 {5.9—7.1, 84.1—101.5}

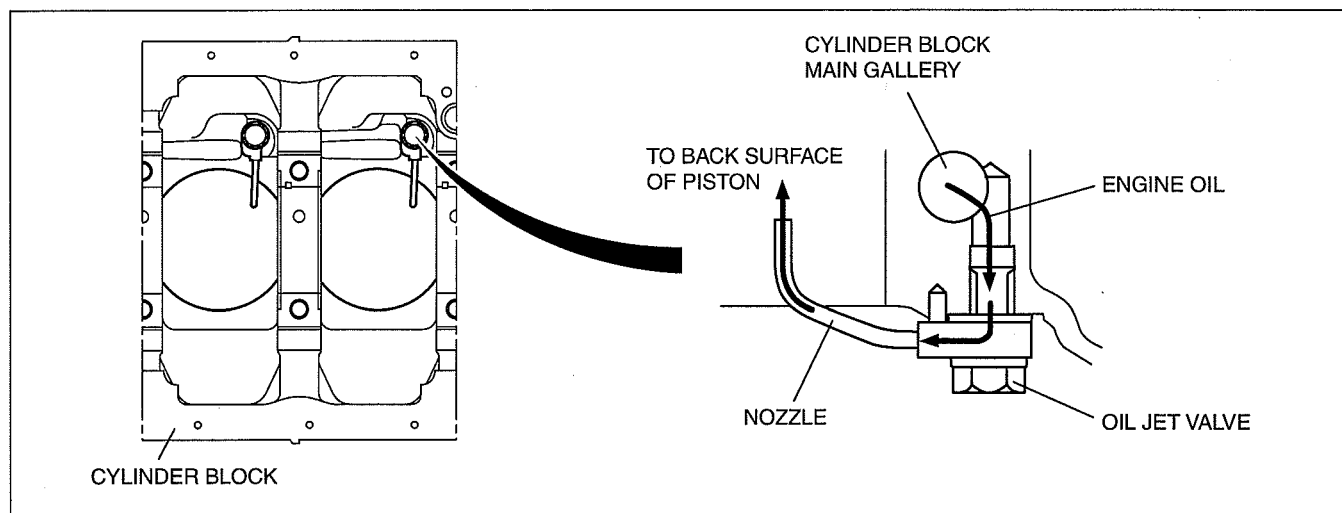
LUBRICATION [WL-C, WE-C]

OIL JET VALVE CONSTRUCTION/OPERATION [WL-C, WE-C]

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Construction

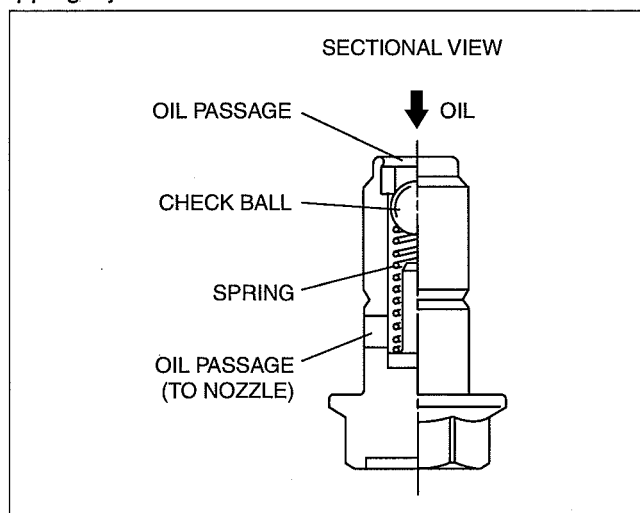
- The oil jet valves are installed in the cylinder block. The oil jet valve nozzles are installed pointed toward the back surface of each piston.
- The oil jet valves are designed to maintain optimum oil pressure in the engine by controlling the oil injection according to the oil pressure applied to the check ball in the oil jet valves.



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Operation

- Oil pressure applied to the check-ball in the oil jet valve opens and closes the oil passage-way to the nozzle and controls oil injection starting and stopping.
- Oil pressure greater than the specified value applied to the check-ball in the oil jet valve opens the oil passage to the spring-pressed nozzle, starting injection. Conversely, oil pressure less than the specified value applied to the check-ball blocks the oil passage by spring force, stopping injection.



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